Evaluating Predictive Skeleton Representations for Neurological Gait Videos

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Abstract

Video-based pose analysis offers a way to study gait outside a motion-capture laboratory, but a classifier can exploit average body configuration or recording conditions without learning useful temporal structure. We evaluate a compact skeleton joint-embedding predictive architecture (S-JEPA) on a collection of videos labeled Normal, multiple sclerosis, or Parkinson's disease. After pose extraction, 88 clips from 41 source videos remain. Five outer folds keep source videos separate, and an inner holdout selects between two label-free training stages before evaluating a frozen representation. We compare the representation with mean pose, landmark visibility, engineered joint-angle features, and a majority-class predictor. Recomputed from saved outer-test predictions, pooled source-weighted macro-F1 is 0.457 for S-JEPA and 0.452 for mean pose; their clip-weighted scores are 0.397 and 0.410. Mean pose performs better in four of five folds. The notebook-reported Random Forest score is 0.411 under source weighting, with a documented feature-construction defect. These comparisons establish no consistent advantage over mean pose in this collection. Unverified participant identities and diagnoses, heterogeneous recordings, and a reflection–mask correspondence problem limit interpretation. A paired source bootstrap gives an S-JEPA-minus-mean-pose difference of 0.005 with a conditional 95% percentile interval of $[-0.160, 0.174]$. *The comparison motivates explicit order-invariant controls before attributing predictive performance to temporal gait information.*

1. Introduction

Changes in walking are relevant to the study of neurological disease, yet their expression varies between people and conditions. Clinical studies have described heterogeneous walking dysfunction in multiple sclerosis (MS), including changes in bilateral coordination [5, 6]. Video-based measurements could make movement analysis more accessible, provided that the relationship between recorded motion, extracted pose, and the intended clinical measurement is established. Validation studies of video-derived gait measures [8, 9] and work on Parkinson's disease (PD) [7] motivate this direction without validating every subsequent classifier trained on video.

Learning from heterogeneous recordings introduces another difficulty. A pose sequence contains body configuration and changes over time, but also measurement effects associated with viewpoint, occlusion, and the pose estimator. Several clips may come from the same source video. A representation can consequently predict a collection label through cues that will not transfer to a new recording setting. Source-separated evaluation and informative simple baselines help assess this possibility, even when the available metadata cannot establish participant independence.

We investigate whether a compact predictive skeleton representation provides a consistent classification advantage over simple pose summaries in a small collection labeled Normal, MS, and PD. The encoder follows the joint-embedding predictive approach developed for images, video, and skeletons [1–3], with stochastic masking of body regions and label-free training on the training sources. A linear classifier then predicts the three collection labels from frozen features.

The empirical contribution is that the learned representation's small pooled lead over mean pose under source weighting reverses under clip weighting and is concentrated in one fold. The evaluation pairs a temporal representation with an order-invariant mean-pose control and a visibility-only control on identical partitions. The observed ordering changes with the evaluation weights, and fold variation is substantial. This makes the study relevant to behavioral phenotyping from markerless pose: it examines what a small video collection can support before the output is interpreted as a clinical measure. It contains no neural recordings or direct neural decoding experiment.

2. Related work

Video-based gait analysis has been evaluated against reference measurements and across clinical populations [8, 9].

Table 1. Collection composition. A padded clip has fewer than 32 retained frames. Windows overlap within a clip and are not independent observations.

Label	Raw	Usable	Sources	Padded	Windows
Normal	26	24	16	5	59
MS	30	29	13	0	171
PD	35	35	12	10	128
Total	91	88	41	15	358

Such studies assess defined movement quantities under specified conditions. Our outcome is instead a supplied video label, with no independently verified diagnosis or reference kinematics. The difference in supervision matters when comparing conclusions across these settings.

Skeleton representation learning includes masked reconstruction and motion pretraining, as illustrated by SkeletonMAE and MotionBERT [12, 13]. I-JEPA and V-JEPA predict learned target features from context [2, 3]; S-JEPA applies a joint-embedding predictive architecture to skeleton sequences [1]. We evaluate a compact implementation with a region-sampling rule motivated by gait anatomy. The study does not isolate the effect of that rule or establish a new general-purpose skeleton-learning method.

Small-sample model evaluation can have large uncertainty, and using the same data for model selection and evaluation can bias the reported error [10, 11]. We therefore separate training, validation, and outer-test sources. This controls source overlap within the specified evaluation procedure, although it cannot account for every decision made during development on this collection.

3. Video collection and pose processing

The collection contains 91 downloaded MP4 clips grouped by an 11-character source-video identifier. The labels Normal, MS, and PD describe collection metadata, not diagnoses established by this study. Pose cleaning retains 88 clips from 41 sources (Table 1). A source identifier is a recording identifier: different sources may contain the same participant, while one source may contain more than one person. Neither possibility is resolved by the available metadata.

The extraction pipeline uses MediaPipe pose estimation from the BlazePose family [4]. It processes full frames and retains the first detected pose. The cached array has 33 landmarks with pixel x, pixel y, and a visibility value for each sampled frame. The pose-estimator interface converts its image-normalized coordinates to pixels before caching; the subsequent body-relative normalization described below is a separate operation. Depth is not used.

Frames are selected with integer stride $\max(1, \mathrm{round}(f_{\mathrm{src}}/15))$. Thus, 15 frames per second is a nominal target rather than an exact common sampling rate. Cleaning requires finite x coordinates for every landmark in at least 30% of sampled frames, trims empty ends, and interpolates missing coordinates and visibility within the sequence. No maximum interior gap length is imposed. At least eight frames must survive cleaning. Three clips are excluded, but the saved exclusion list does not identify a separate failure criterion for each.

For pixel position $x_{tj} \in \mathbb{R}^2$, hip midpoint h_t, and shoulder midpoint s_t, the normalized coordinates are

$$\widetilde{x}_{tj} = \frac{x_{tj} - h_t}{\max(\|s_t - h_t\|_2, 10^{-3})}. \quad (1)$$

Spatial normalization leaves the already cleaned visibility values unchanged. This transformation removes absolute pelvis translation and apparent torso size. Within each frame it preserves defined angles under a common positive scale, but frame-varying scale changes displacement over time. The resulting trajectories are not calibrated measurements of physical speed or stride length.

Each source is assigned to a partition before generating 32-frame windows with stride 16. A shorter clip is padded by repeating its final frame. There are 358 windows in total, including 15 padded clips. Padding and interpolation can change apparent dynamics; their distribution across labels is therefore reported explicitly.

4. Predictive skeleton representation

4.1. Stochastic region masking

A training window contains all 33 landmarks and three channels. Four consecutive frames for one landmark form a token, giving eight time blocks and $8 \times 33 = 264$ tokens. The masking sampler operates on seven overlapping regions: head, mouth, left arm, right arm, trunk, left leg, and right leg. It draws one region and a contiguous span of one to six time blocks, adds their tokens to the target set, and repeats until it reaches the rounded target budget of 158 tokens (nominally 60%) or a 36-draw guard. Span length and admissible start position are drawn uniformly. Region overlap and the final draw mean the realized target fraction can exceed 60%.

Twelve landmarks influence region-selection weights: both shoulders, hips, knees, ankles, heels, and foot indices. A region containing any of these landmarks receives a factor of 1.5, divided by the average number of regions containing its constituent joints; the resulting weights are normalized over regions. This is a bias in region selection. It neither fixes which joints are masked nor makes each highlighted joint 1.5 times as likely to be hidden. Shoulders and hips appear in two regions, so the overlap correction does not equalize their final masking frequencies.

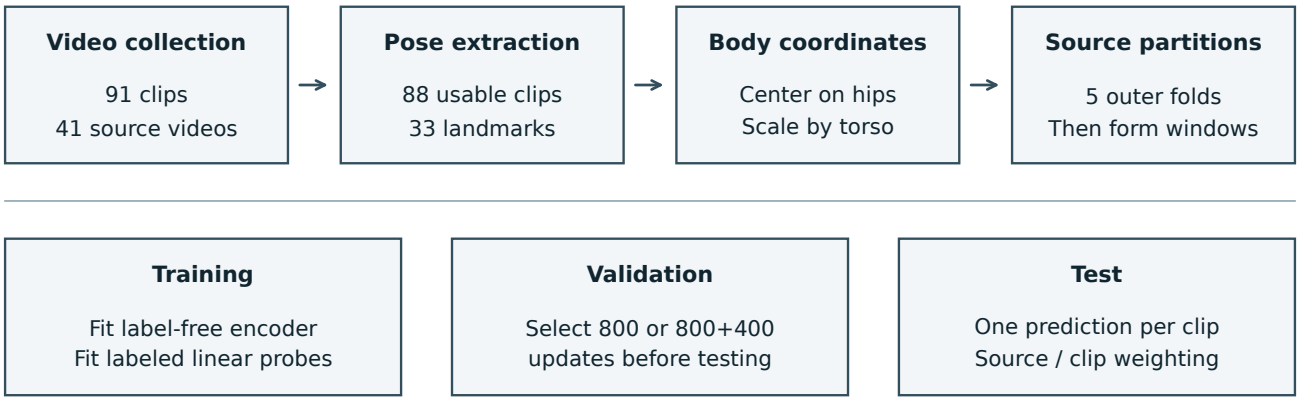


Figure 1. The notebook-defined evaluation. Pose extraction precedes body-relative normalization. Source partitions are fixed before window generation; encoder training and probe fitting use only training sources, and validation chooses the training stage. Test predictions are pooled under two explicitly defined weighting rules.

The arms also receive the boost because they contain shoulders, so their elbows, wrists, and hand landmarks inherit that region weight. Every example receives a new mask. The sampler guarantees at least one visible token from the twelve-landmark set somewhere in the window, as well as nonempty target and context sets. The retained token can be a shoulder: neither hip nor leg context is guaranteed even somewhere in a particular window, and an entire time block can be hidden. In the notebook's separate seed-zero bank of 512 masks, the mean target fraction is 63.3%, and every joint is visible somewhere in at least 74.0% of windows. These sampler diagnostics describe draws from the rule, not realized exposure in the five-fold training run. The anatomical choice is a design hypothesis motivated by gait analysis. No comparison with uniform, motion-dependent, or alternative anatomical masking establishes its benefit.

4.2. Encoder and training objective

The tokenizer maps four frames $\times$ three channels for one landmark (12 numbers) linearly to 96 features, then adds learned joint and time embeddings. The context encoder is a three-layer transformer of width 96 with four attention heads, feed-forward expansion two, and no dropout. The two-layer predictor also has width 96 and adds learned joint and time positions so that hidden slots remain distinguishable. A target encoder starts as a copy of the context encoder and tracks it through an exponential moving average (EMA).

The student receives a geometrically transformed window. Rotation is sampled within $\pm 15°$, scale within $[0.9, 1.1]$, and translation within ± 0.1 body-normalized units per axis. Horizontal reflection occurs with probability 0.5 and exchanges left and right landmark slots, including their visibility values. One transform applies to the whole window. Within the context encoder, target positions are excluded as attention keys and values; their output vectors are replaced by a shared learned mask token before the predictor receives them. The per-example implementation retains computations at hidden query positions rather than physically dropping those positions. The target encoder receives the complete, untransformed window without a loss gradient. The loss compares predicted and target distributions over the 96 feature dimensions at masked slots; these distributions are not class probabilities.

With predicted features u, target features z, and running center c, the distributions are $p = \mathrm{softmax}(u/0.10)$ and $q = \mathrm{softmax}((z - c)/0.06)$. Cross-entropy is averaged over masked tokens within each example and then over examples. Once per batch, the center retains 90% of its previous value and adds 10% of the masked teacher-feature mean. This center is token-weighted, whereas the loss weights examples equally after their within-example averages. EMA momentum follows a cosine schedule from 0.996 toward one.

Training samples windows with replacement so that each training source has equal total probability. The batch size is 32, with 800 initial updates and a 400-update continuation on the same training sources. Both stages use the same label-free objective across all three labels; neither a normal-only pretraining restriction nor a class-aware VICReg loss is active. Continuation retains network weights but resets the optimizer, center, random streams, and schedules. It therefore differs from uninterrupted training for 1,200 updates.

AdamW uses peak learning rate 10^{-3}, weight decay 0.05, coefficients $(0.9, 0.95)$, gradient clipping at norm one, and linear warmup over the first 10% of updates followed by cosine decay. Initialization and training use seed 42. These settings define the evaluated configuration rather than an independently optimized architecture.

Table 2. Source/clip counts in the fixed partitions.

Fold	Training	Validation	Test
0	24/51	8/19	9/18
1	24/58	9/17	8/13
2	24/54	9/17	8/17
3	24/53	9/21	8/14
4	24/45	9/17	8/26

4.3. Frozen representation and classifier

At readout, the frozen EMA target encoder receives every token of the complete window. We average its features at the 161 token positions selected by one fixed mask drawn with seed zero, then average window vectors within each clip. The mask here selects output positions; it does not hide the corresponding input coordinates. All 33 landmarks occur at least once in the selected positions. The result is one 96-dimensional vector per clip.

A training-fitted standardizer and class-balanced logistic regression ($C = 1$, maximum 2,000 iterations, seed 42) form the classifier. Each training stage has its own classifier. Validation source-weighted macro-F1 selects the stage, with ties retaining the initial stage; the chosen model is then applied to outer-test clips. The scaler, classifier, and encoder are not refit on combined training and validation sources. Class balancing in the linear head does not make its training loss source-balanced.

5. Evaluation

5.1. Partitions and baselines

Label-stratified five-fold splitting operates on one row per source, then expands source assignments to clips. Within each outer development pool, the first split from a four-fold label-stratified source splitter forms the validation holdout. The outer seed is 42 and the inner seed is $43 + k$ for fold k. This gives five outer evaluations with one inner holdout each, rather than an exhaustive nested search. Every retained clip has exactly one outer-test prediction.

Mean pose averages normalized x, y over each complete clip, yielding 66 features without temporal order. Visibility uses the mean and standard deviation of each landmark's visibility, also yielding 66 features. Both use the same linear-classifier recipe as the learned representation. Mean pose retains posture and viewing geometry, while visibility may reflect occlusion, body orientation, or detector behavior. Neither is a pure measurement of recording conditions.

The engineered-feature baseline reads cleaned pixel coordinates. Its intended vector contains six per-side hip, knee, and ankle angle means, three absolute bilateral mean differences, and six per-side ranges. The legacy container has 82 fields, but only 15 columns vary in each training fold along this extraction route. In the notebook-reported experiment, the left-ankle range duplicates the right-ankle range. This omits one intended measurement and limits conclusions about the intended baseline. Its Random Forest has 100 trees, maximum depth five, square-root feature subsampling, class-balanced training, and seed 42. Columns constant in the training partition are removed, and the remaining features are standardized using that partition. A majority-class baseline predicts the most frequent training-clip label in each fold. All methods use identical outer partitions, but the S-JEPA procedure additionally selects between two stages on validation data.

5.2. Scoring

Our primary descriptive measure is pooled source-weighted macro-F1. If source s contributes n_s clips, each clip receives weight $1/n_s$. For supplied label a and prediction b, weighted confusion counts are

$$C_{ab} = \sum_i \frac{\mathbf{1}(y_i = a, \widehat{y}_i = b)}{n_{s(i)}}. \tag{2}$$

The class F1 score is $2C_{aa}/(\sum_b C_{ab} + \sum_b C_{ba})$, and macro-F1 averages the three class values. Each source contributes total weight one before scoring. This is not an average of source-level F1 scores or a classification of one aggregated prediction per source.

We also report pooled clip-weighted macro-F1 and the mean and population standard deviation of the five source-weighted fold scores. Pooled F1 differs from the mean of fold F1 because the score is nonlinear. Fold standard deviation describes heterogeneity and is not a confidence interval; the folds have overlapping training sets. We add a retrospective paired source bootstrap to the notebook-associated outer-test predictions. In each of 10,000 draws (seed 20261004), we sample 16 Normal, 13 MS, and 12 PD sources with replacement within label strata. Each sampled occurrence contributes total weight one, distributed equally among that source's clips, and every method uses the same source draws. We recompute pooled macro-F1 and paired differences, reporting the 2.5th and 97.5th percentiles.

These are conditional source-resampling intervals with class-source counts and fitted predictions held fixed. They exclude retraining, seed, split, stage-selection, and development uncertainty. Sources are treated as independent clusters even though participant overlap is unresolved, and shared training data induce dependence across fitted outer-fold models. We use the intervals as descriptive sensitivity summaries, not a confirmatory population test or evidence of equivalence. This analysis was added for the manuscript; it is not a saved notebook output.

Table 3. Held-out macro-F1. Source and clip columns pool all outer-test predictions; the last column summarizes source-weighted fold scores. SD is descriptive. *The notebook RF contains a duplicated ankle-range feature.

Representation	Source	Clip	Fold mean ± SD
S-JEPA	0.457	0.397	0.438 ± 0.196
Mean pose	0.452	0.410	0.427 ± 0.132
Joint angles + RF*	0.411	0.397	0.395 ± 0.098
Visibility	0.318	0.318	0.285 ± 0.104
Majority class	0.259	0.202	0.159 ± 0.022

6. Results

Table 3 reports the full-budget experiment described in the capstone notebook, with scores checked against its 88 saved outer-test rows. S-JEPA reaches source-weighted macro-F1 0.457 and mean pose reaches 0.452, a difference of 0.0047. Under clip weighting their ordering reverses: mean pose reaches 0.410 and S-JEPA 0.397. Mean pose correctly classifies 37 clips and S-JEPA 36. One MS source contributes 13 clips, making the choice of weighting consequential.

The conditional interval for S-JEPA macro-F1 is $[0.329, 0.579]$. Its paired difference from mean pose is 0.0047 with interval $[-0.160, 0.174]$; against the notebook RF it is 0.0454 with interval $[-0.133, 0.220]$. Both span zero. The data and this conditional procedure leave the directions uncertain without showing that the methods are equivalent.

Mean pose exceeds S-JEPA in four of five outer folds (Figure 2a). The pooled difference consequently does not describe a consistent foldwise improvement. The visibility-only score is lower than both coordinate-based representations, but this comparison cannot quantify how much recording information remains in their features. Mean pose is a demanding control here because it retains body configuration while discarding frame order. S-JEPA fold scores range from 0.217 to 0.780. Its raw correct counts are 10/24 Normal, 8/29 MS, and 18/35 PD clips; source-weighted recalls are 0.396, 0.429, and 0.569 (Figure 2b). The average score therefore coexists with substantial errors in all three labels.

The initial stage is selected in folds zero through two, and continuation in folds three and four. Continuation lowers the final sampled training-minibatch loss in every fold, yet its validation score improves in only two. This shows a mismatch between those optimization summaries and stage selection; it does not compare the expected losses of both stages on a common evaluation set.

The top-level notebooks define the reported experiment. The capstone notebook contains a written result table and executable evaluation procedure, but its code cells have no saved outputs. We verify its written results against the corresponding retained evaluation artifacts. The companion audit record (`evidence/reproducibility.md`) names the notebook-associated execution and separately identifies the corrected verification. It records input hashes and the command for `evidence/results_audit.py`, which recomputes every table metric and interval into `results-audit.json`. The original PDF is not an evidence source when its numbers disagree with those records.

6.1. Separate verification of the RF correction

A later saved evaluation uses the true left-ankle range in place of the duplicated right-ankle range. As a manuscript-stage verification, we refit the small deterministic RF baseline on the existing partitions. Current extraction reproduces all 88 later predictions, and replacing only that feature with the historical duplicate reproduces all 88 notebook-associated RF predictions. Twelve predictions change between versions. This identifies the feature correction as sufficient to reproduce the RF difference under these settings, while preserving both records.

The corrected RF yields source-weighted macro-F1 0.325 and clip-weighted macro-F1 0.321, compared with 0.411 and 0.397 in the notebook-reported implementation. S-JEPA minus corrected RF is 0.1314 with conditional interval $[-0.046, 0.304]$. A correction can lower performance; the direction alone says nothing about which implementation faithfully computes the intended feature. This verification does not establish superiority over a well-tuned classical gait system. Both saved executions use the same data, split, and training seed, and their S-JEPA hard predictions coincide despite different embedding values. They are not independent-seed replications.

7. Discussion and limitations

The central comparison gives little evidence of a consistent advantage over mean pose under this configuration. A small positive difference under one weighting and a negative difference under another leave the practical ordering sensitive to how recordings are represented in the score. They do not establish equivalence, prove that temporal information is unnecessary, or identify the information used by the encoder. An untrained-encoder baseline and controlled changes to temporal order would address different parts of that question.

The masking and augmentation implementation also restricts interpretation. Reflection swaps landmark slots in the student view, but the target mask and the unreflected teacher slots retain their original indices. A teacher target joint can therefore remain visible as transformed input in the student's opposite-side context when the unpermuted mask treats that slot as context. This occurs inside training examples rather than through test-source overlap. Its effect on the learned representation has not been measured, and

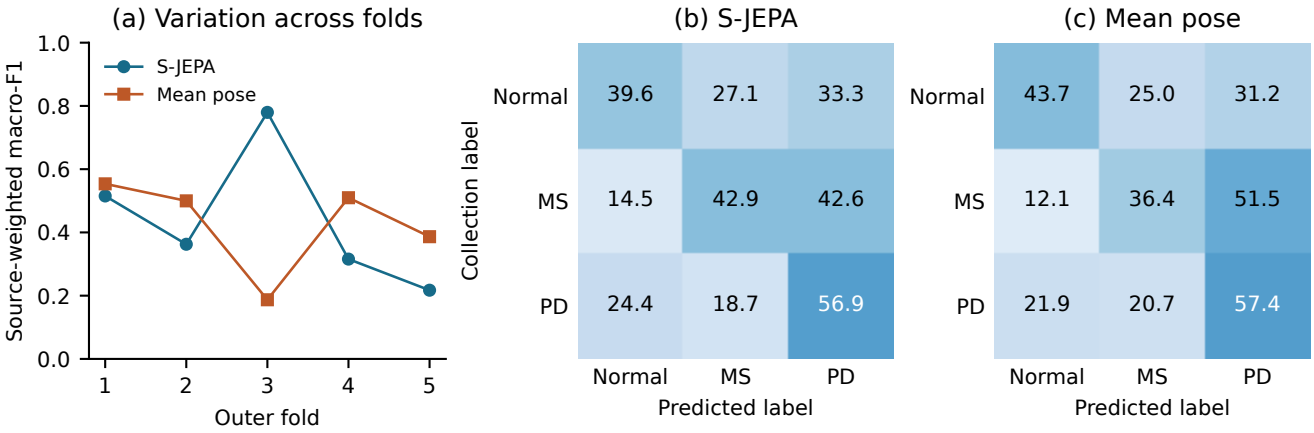


Figure 2. Notebook-associated outer-test results. (a) Source-weighted macro-F1 by outer fold; connecting lines identify the methods across fixed folds and imply no time trend. (b,c) Row-normalized source-weighted confusion percentages. Rows are collection labels, columns predictions, and each source contributes total weight one before row normalization. These are not percentages of clinically verified patients.

the present results should not be treated as a clean test of anatomically aligned masked prediction.

The collection is small and heterogeneous. Source grouping prevents clips with the same recorded identifier from crossing partitions, but unknown identity overlap can remain. The supplied labels are not verified clinical diagnoses. Pixel-to-body normalization removes some information while retaining viewpoint effects; interpolation and repeated-frame padding further alter the temporal input. Mean-pose and visibility controls expose useful alternative predictors without exhausting potential shortcuts.

The available study records do not establish participant consent for secondary health inference, a complete source-media rights assessment, or an ethics-review determination. Public accessibility alone does not settle those questions. The manuscript therefore makes no claim that these records authorize unrestricted redistribution, and the classification results do not support individual clinical decisions. Any release or follow-up clinical study requires a documented basis for using the recordings and their associated labels.

The next empirical priorities follow from the measured uncertainties: correct or explicitly control reflection correspondence, compare trained and untrained encoders under identical readout rules, and test temporal order while keeping the evaluation groups fixed. A larger collection with verified participant identities and diagnoses would then permit participant-separated evaluation and external validation. Calibration and transfer to new sites remain untested.

8. Conclusion

Source-separated evaluation of 88 gait clips provides a bounded test of a compact predictive skeleton representation. The learned features and mean pose achieve similar pooled source-weighted scores, with mean pose favored under clip weighting and in four of five folds. This motivates stronger temporal and acquisition controls before interpreting classification as evidence of clinically meaningful gait representation. The useful result is the comparison and its limits; a robust clinical measure requires independent validation beyond these source-video labels.

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